Limit Order Book Mid-Price Forecasting by Ensemble Learning

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Introduction Motivation

- **Challenge:** Predicting short-term price movements in high-frequency markets (LOB) is difficult due to:
  - Low signal-to-noise ratio.
  - Non-stationary nature of financial time series.
  - Complex non-linear spatiotemporal dependencies.

- **Research Gap:**
  - *Linear Models:* Interpretable but fail to capture non-linearities.
  - *Deep Learning:* Powerful but prone to overfitting on stochastic market data.

- **Proposed Solution:** A comprehensive hybrid framework integrating **Ensemble Learning** (Lasso, XGBoost, and Deep Neural Networks).

Dataset Description

High-Dimensional LOB Data

- **Training Set:** 23,532,026 observations.
- **Testing Set:** 4,706,406 observations (OOS).
- **Structure:** 8 levels of Ask/Bid.
- **Features per level:**
  1. Rate (Price)
  2. Size (Volume)
  3. **Nc (Order Count)**

Target Variable (y)

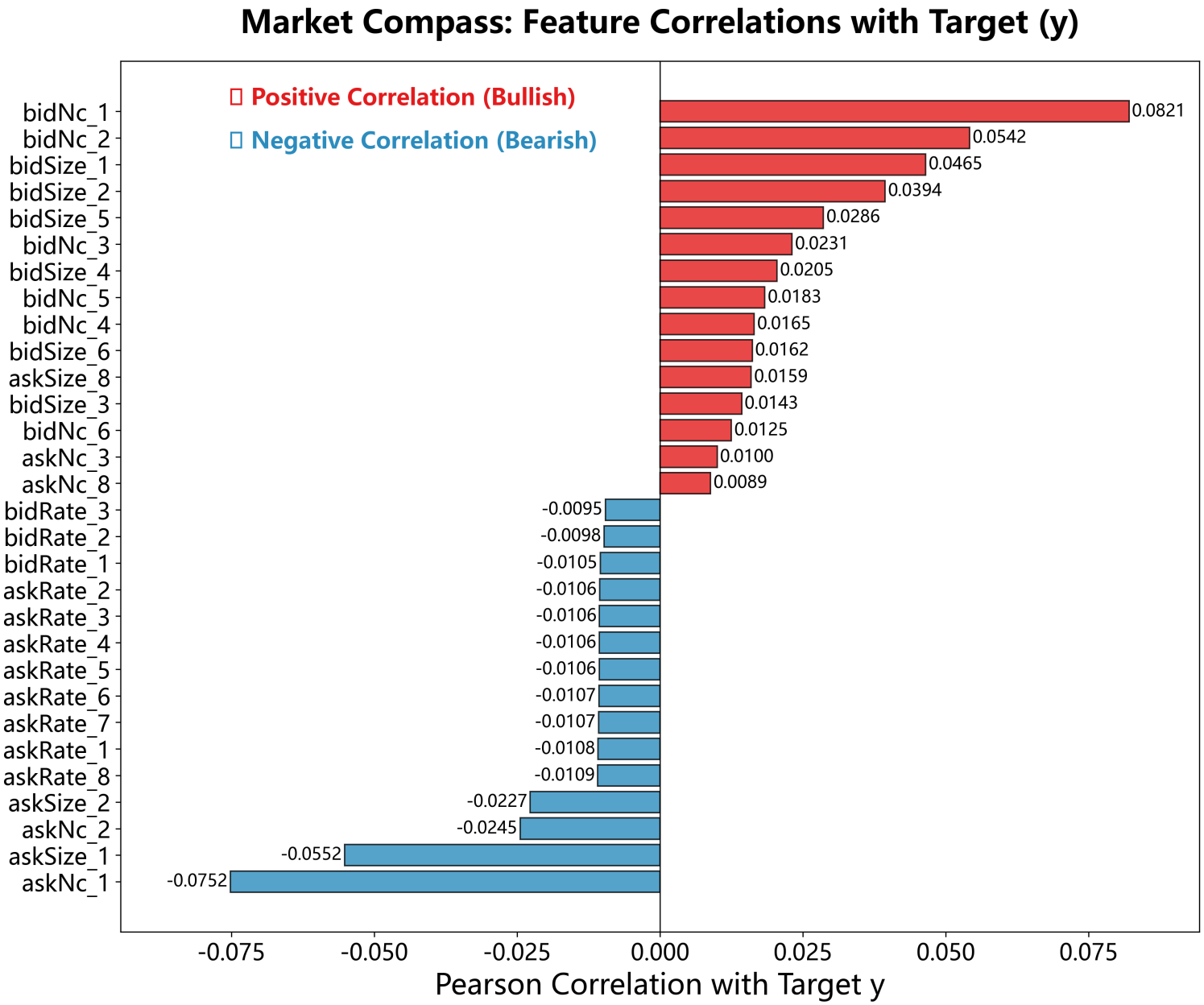
- Future return of the mid-price over a fixed horizon.

Exploratory Data Analysis (EDA)

- **Key Insight 1: Predictive Superiority of Order Density**
  - The number of orders (N_t) consistently shows higher correlation with returns than total volume (V_t).
  - High density implies broader consensus; large volume may be iceberg orders.

- **Key Insight 2: Microstructure Anomalies**
  - *Non-monotonicity:* Deep ask levels (e.g., Level 8) show positive correlation with price increases (counter-intuitive).
  - Likely a proxy for strategic behaviors like **Spoofing** or Layering.

EDA: Feature Correlations



Feature Engineering

Constructed a 19-dimensional feature vector focusing on microstructure properties:

1. **Price Momentum:** Multi-scale rolling momentum (30, 60, 120 ticks).
2. **Order Flow Imbalance (OFI):**
   - Introduced **Count-based Imbalance** (I_t^{cnt}) alongside Volume-based.
   - Optical Imbalance (ϕ_t): Concentration of liquidity at best quotes.
3. **Market Dynamics:**
   - Net Order Flow (NOF): Proxy for aggressive market orders.
   - Market Attack Intensity: Detecting Buy/Sell attacks.
4. **Spoofing Proxy (S_t):**
   - Ratio of "Far Depth" (levels 5-8) to "Near Depth" (levels 1-3).
   - Captures fake liquidity intent.

Methodology: Multi-Tier Architecture

A system ranging from linear baselines to deep neural networks.

1. **Linear Baseline (Lasso):**
   - GPU-accelerated sparse modeling.
   - Acts as an interpretability anchor.
2. **Non-Linear Interaction (XGBoost):**
   - Uses ‘gpu_hist‘ method.
   - Captures discrete threshold effects and tabular features.
3. **Hybrid Neural Network (Magic-Mixer):**
   - **Backbone: MLP-Mixer** (Token Mixing + Channel Mixing) to extract spatiotemporal dependencies without self-attention overhead.
   - **Gated Residual Learning:** Fuses linear signals with deep residuals.

Adaptive Gated Linear Fusion

To prevent overfitting and ensure stability, the Hybrid NN uses a Dual-Pathway mechanism:

$$\hat{y} = (1 - \sigma(w_{gate})) \cdot \Phi_{deep}(\mathbf{X}) + \sigma(w_{gate}) \cdot \mathbf{W}_{lin}^T \mathbf{x}_{lin} \tag{1}$$

- $\Phi_{deep}(\mathbf{X})$: Output of MLP-Mixer.
- $\mathbf{x}_{lin}$: Canonical linear features (Momentum, Imbalance).
- w_{gate}: Learnable parameter balancing linear reliability and non-linear expressiveness.

Single Model Performance

Results on Out-Of-Sample (OOS) data:

Model	**MSE**	**Correlation**	R^2
Lasso	4.4278	0.0194	-0.0051
Hybrid NN	4.5727	0.0444	-0.0380
XGBoost	**4.3213**	**0.1451**	**0.0190**

- **Lasso:** Fails to capture alpha ($R^2 < 0$), proving non-linearity is essential.
- **XGBoost:** Dominates single models, robust to discrete microstructure features.

Ensemble Optimization Results

Combining models significantly enhances performance:

- **Strategy:** Optimized weighted ensemble.
- **Weights:**
  - XGBoost: 61.6%
  - Hybrid NN: 38.4% (Captures orthogonal temporal signals)
  - Lasso: 0.0%

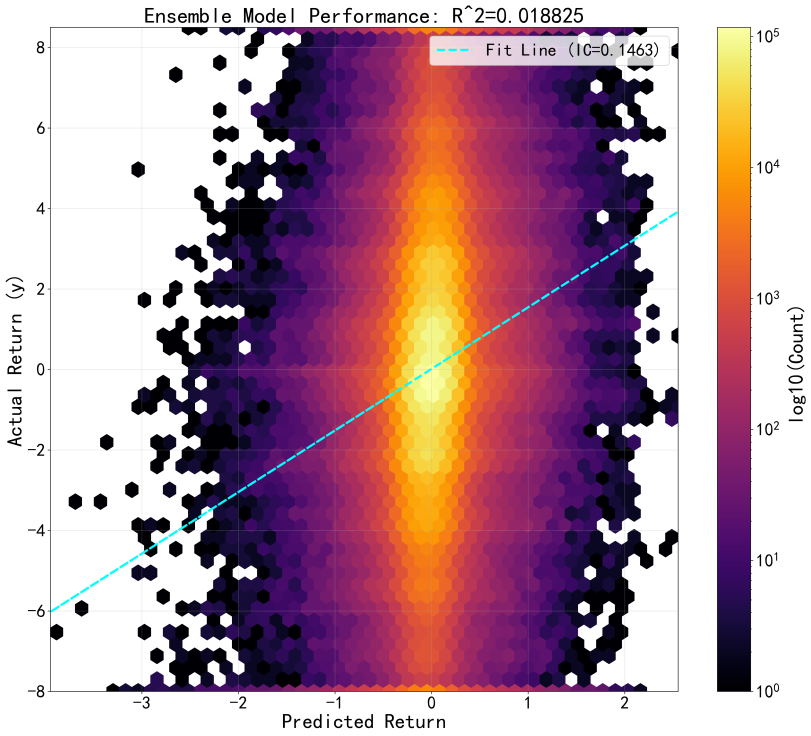
Final Result

Correlation: **0.1623**

Improvement: **+10.63%** over the best single model.

Prediction Visualization

- **Hexbin Density Plot:** Prediction vs. Actual Returns.
- The clear positive slope (Cyan line) confirms robust directional forecasting capabilities.



[Insert Figure 2 here]

Conclusion

1. **Data Engineering:** Confirmed that Order Count (N_t) and depth-based Spoofing proxies are critical for LOB forecasting.
2. **Architectural Innovation:**
   - MLP-Mixer efficiently extracts spatiotemporal features.
   - Gated Residual Learning stabilizes training in low SNR regimes.
3. **Robust Paradigm:**
   - The ensemble of Gradient Boosting and Deep Neural Networks yields industrial-grade alpha.
   - Achieved substantial performance gain (+10.63%).

Q & A